

Midterm Sample

ECON 441: Introduction to Mathematical Economics

Instructor: Div Bhagia

Print Name: _____

This is a closed-book test. You may not use a phone or a computer.

Time allotted: 70 minutes

Total points: 20

Please show sufficient work so that the instructor can follow your work.

I understand and will uphold the ideals of academic honesty as stated in the honor code.

Signature: _____

1. (5 pts) Answer the following questions.

(a) (1 pt) Consider a mapping $f(x)$. For two distinct values of x , x_0 and x_1 , $f(x_0) = f(x_1)$. Is f a valid function? Answer yes or no.

(b) (1 pt) Consider two sets A and B , where A is the set of all odd real numbers and B is the set of all even real numbers. What is the union of A and B ?

(c) (1 pt) Find the union and intersection for the following sets:

$$A = \{x : x \text{ is an even number}\} \quad B = \{2, 4, 8\}$$

(d) (1 pt) Does the inverse of the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ exist? If yes, find the inverse.

(e) (1 pt) Consider the following two-variable function:

$$f(x, y) = x + y$$

where $x \in (0, 1)$ and $y \in (0, 1)$. What is the range of f ?

2. (5 pts) Consider the following matrix

$$A = I - X(X^T X)^{-1} X^T$$

(a) (3 pts) Is A a square matrix? Show your work or reasoning that led you to this conclusion.

(b) (2 pts) Prove that A is idempotent i.e. $AA = A$.

3. (10 pts) Two types of cars, gasoline (g) and electric (e), are available in the market. Denote the price of gasoline cars by p_g and the price of electric cars by p_e , and the corresponding quantities by q_g and q_e . The supply and demand equations are as follows:

Supply Equations:

$$q_e^s = 30 + p_e$$

$$q_g^s = 20 + p_g$$

Demand Equations:

$$q_e^d = 100 - 2p_e + p_g$$

$$q_g^d = 50 + p_e - p_g$$

Note: At equilibrium, supply is equal to demand: $q_e^s = q_e^d = q_e$ and $q_g^s = q_g^d = q_g$.

- (a) (2 pts) Using the equilibrium conditions, derive two equations in the two unknown prices p_e and p_g . Rearrange each equation so that all terms with variables are on one side and all constants are on the other.

- (b) (2 pts) Express this system of equations in matrix format as $Ax = b$. Clearly specify what A , x , and b are.

- (c) (2 pts) Calculate the adjoint of A .
- (d) (1 pt) Calculate the determinant of A . Is A nonsingular?
- (e) (1 pt) If you premultiply A^{-1} on both sides of the equation $Ax = b$, you should be able to derive an expression to solve for x . Write down this expression.
- (f) (2 pts) Using the expression in (e), solve for the equilibrium prices p_e^* and p_g^* .